

Silicon particle size analysis - Hackathon-Polaron batches

Answer: No per-image metric distinguishes the batches at the 5% level (all Kruskal-Wallis $p > 0.05$): the batches cannot be statistically distinguished on silicon-particle size with these data. Pooled number-weighted diameters (all images): Batch_1: D50 0.42 μm [0.40-0.52], D90 1.31 μm [0.97-2.00], $n=6398$ from 7 images; Batch_2: D50 0.48 μm [0.44-0.51], D90 1.87 μm [1.59-2.15], $n=2939$ from 7 images; Batch_3: D50 0.47 μm [0.44-0.53], D90 1.66 μm [1.36-2.07], $n=8767$ from 17 images. Because every measured particle is micron-scale while the paper studied 3-27 nm particles, only the paper's qualitative trend (larger silicon units $\rightarrow$ more mechanical damage $\rightarrow$ shorter CYCLE life; no size effect on CALENDAR life) is invoked - its quantitative results do not apply here. Assumption: bright particles are silicon-based, grey bulk is graphite, from image appearance only.

Assumption throughout: bright BSE particles are taken to be silicon-based and the grey bulk to be graphite, from image appearance only (no EDS confirmation).

Method

BSE images only (31 total: Batch_1=7, Batch_2=7, Batch_3=17). Grey levels taken as the per-pixel median of the three identical RGB channels; no information banner was present in any image. Each image was normalised to its 1st-99.5th percentile range, Gaussian-smoothed (sigma 1.5 px), and thresholded for bright particles at the histogram valley between the bulk peak and a bright peak; images without a separate bright peak used the upper threshold of 3-class multi-Otsu (flagged in the data). Masks were opened (disk $r=2$), objects <150 px dropped, holes filled, and touching particles split by distance-transform watershed. Particles touching the image border are excluded from all size statistics. Pixel size is ~ 25.0 nm/px in every image (TIFF resolution tags). Statistics treat the IMAGE as the independent unit: Kruskal-Wallis plus pairwise Mann-Whitney with Holm correction on per-image medians/D90/max/large fraction, and 95% bootstrap CIs resampling images. The 'large' cut-off is the pooled D90 across all images - a relative limit, since the paper offers no physical cut-off applicable at our sizes.

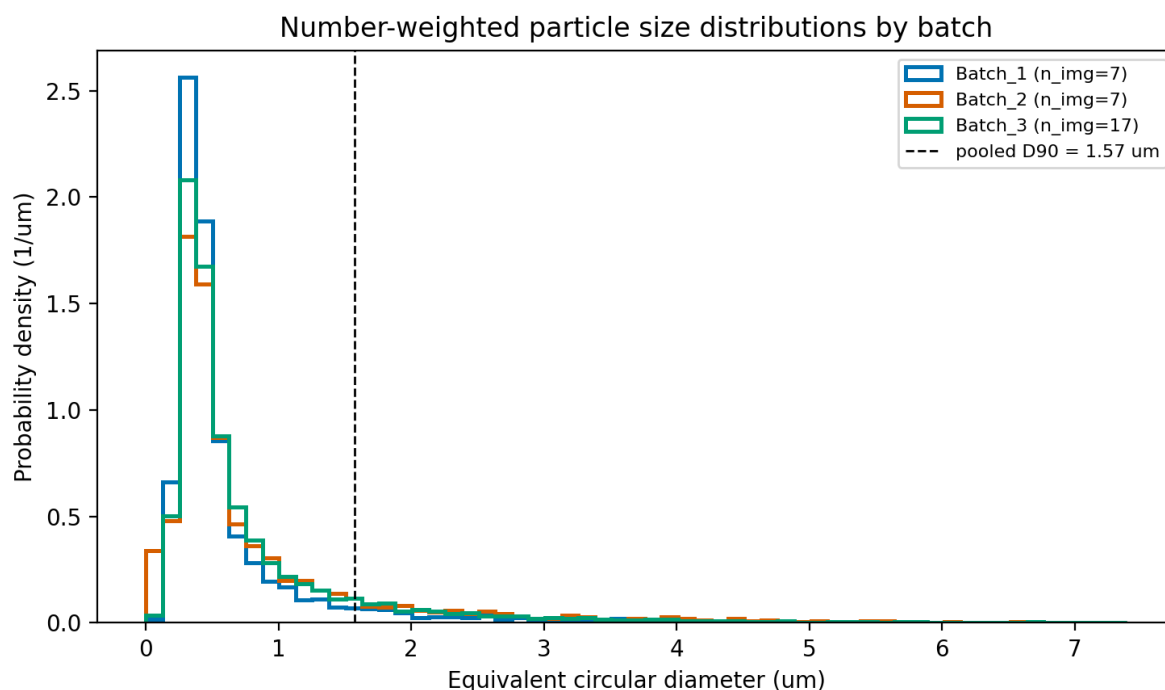


Fig. 1 - Number-weighted diameter density per batch; dashed line is the pooled-D90 'large' cut-off.

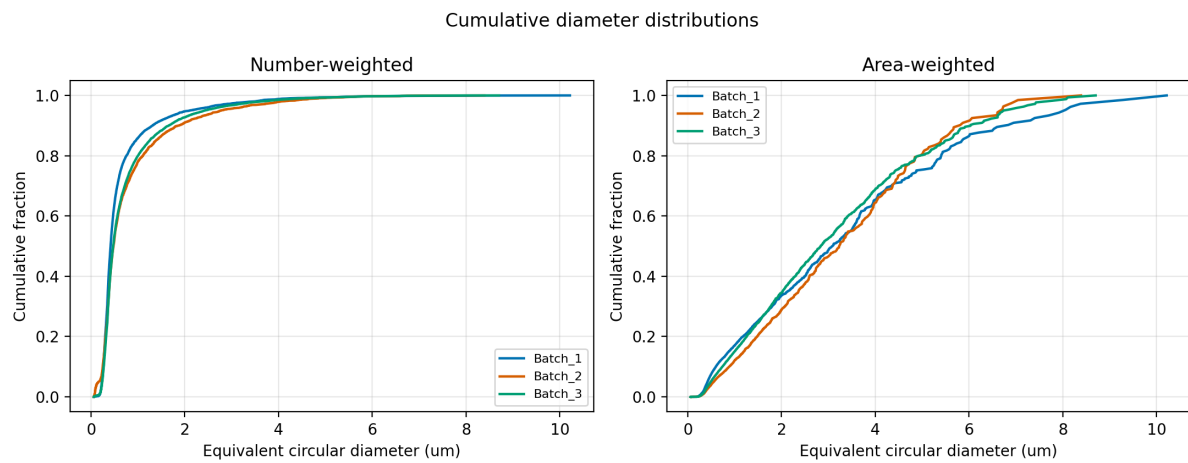


Fig. 2 - Cumulative diameter distributions, number-weighted (left) and area-weighted (right); a few large particles dominate the silicon volume.

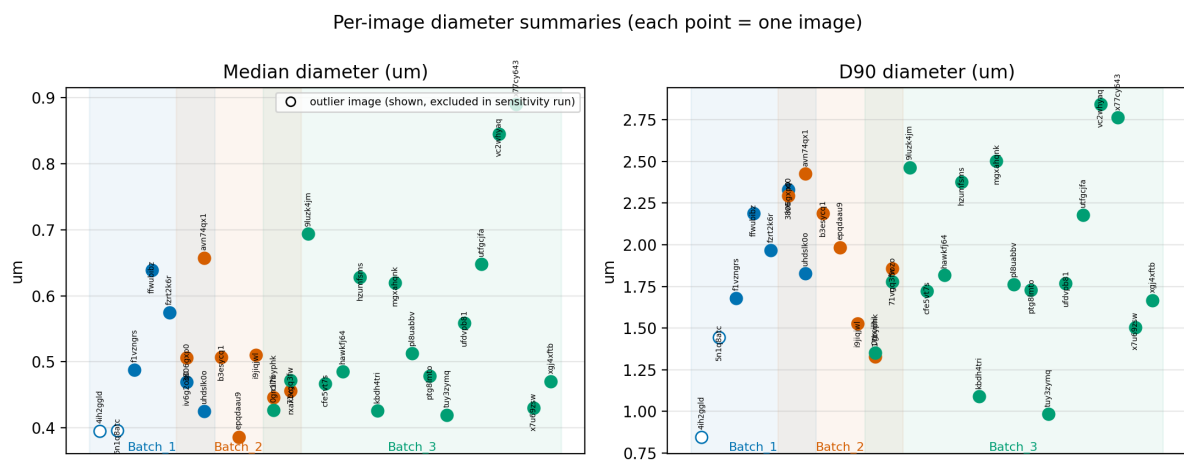


Fig. 3 - Per-image median and D90 grouped by batch; open markers are the two low-contrast outlier images.

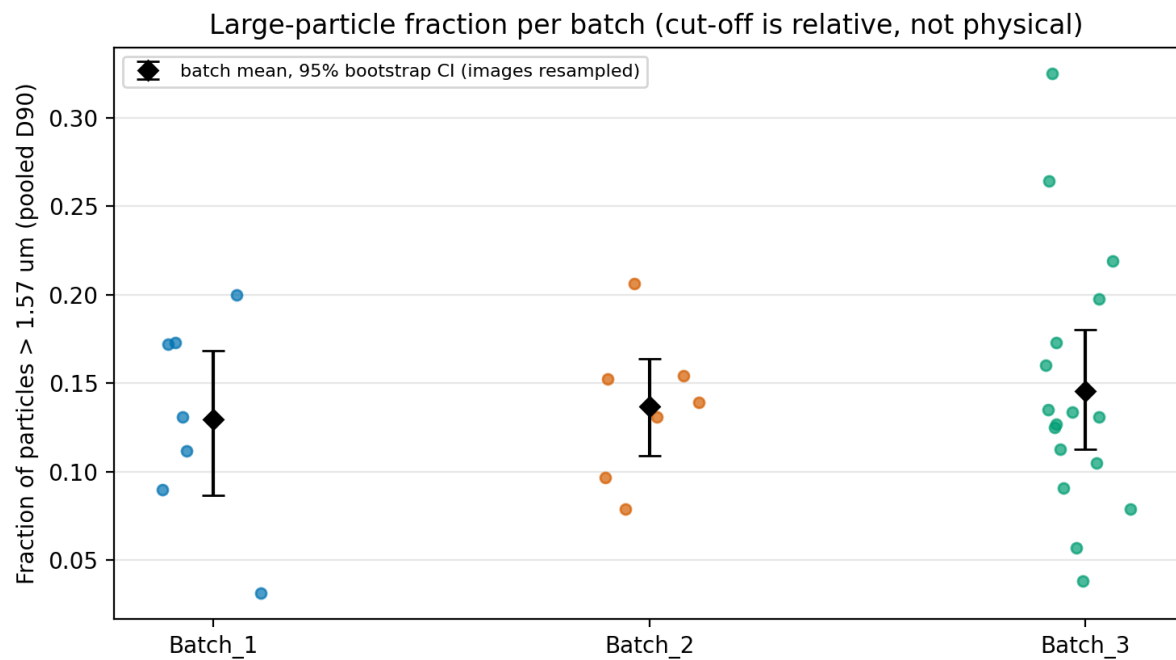


Fig. 4 - Fraction of particles above the pooled-D90 cut-off per image, with batch means and bootstrap 95% CIs.

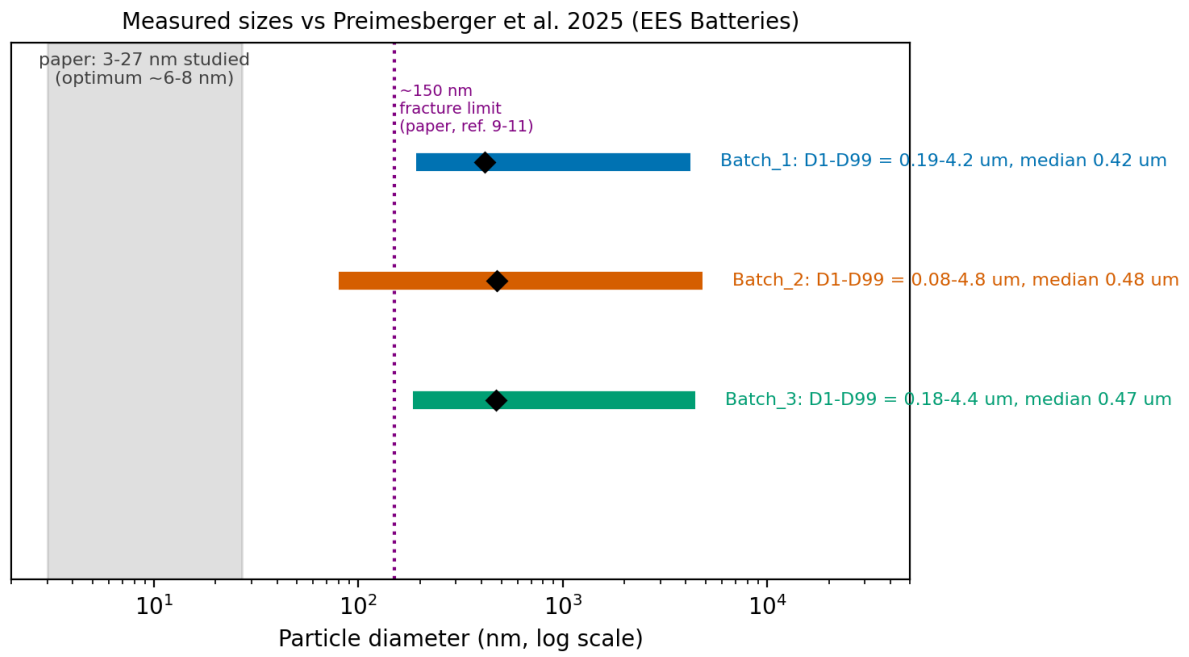


Fig. 5 - Measured size range per batch vs the 3-27 nm range studied in the paper and the ~150 nm fracture threshold it cites.

Batch statistics

batch	images	particles	area frac	D50 um	D50 95%CI	D90 um	D90 95%CI	D99 um	max um	frac>cut
Batch_1	7	6398	0.081	0.42	[0.40,0.52]	1.31	[0.97,2.00]	4.20	10.22	0.130
Batch_2	7	2939	0.055	0.48	[0.44,0.51]	1.87	[1.59,2.15]	4.83	8.39	0.137
Batch_3	17	8767	0.061	0.47	[0.44,0.53]	1.66	[1.36,2.07]	4.43	8.70	0.145

What the paper says, and whether it applies

Confirmation: DOI 10.1039/d4eb00020j resolves to EES Batteries, 2025, vol. 1, pp. 298-309: 'The effect of nanoparticle size on calendar and cycle lifetimes of silicon anode lithium-ion batteries', Preimesberger et al. (NREL).

Sizes studied: "Silicon nanoparticles ranging from 3-27 nm in diameter were made using a plasma-enhanced chemical vapor deposition (PECVD) process" (Methods, p. 299). Ten sizes were tested; the conclusions name 6-12 nm as the better-performing range and ~6-8 nm as optimal (Results p. 303, Fig. 1; Conclusions p. 307).

How size was measured: "Average particle diameters were confirmed by X-ray diffraction, shown in Fig. S1" (Methods, p. 299).

Electrode composition: Pure high-silicon anodes, NOT a silicon-graphite blend: "Slurries were made with 89 wt% Si ... 1 wt% single-walled carbon nanotubes (SWCNTs), and 10 wt% polyimide binder (P84)" (Methods, p. 300). Our sample is believed to be a graphite electrode containing silicon-based particles - a different system.

Size vs cycle life: "In general, as particle size decreases, the capacity retention increases, resulting in better cycle performance" (Results, p. 301-302, Fig. 1a). "Particle diameter, cycle life, and coulombic efficiencies are all correlated with each other" (p. 306, Fig. 5).

Size vs calendar life: "The rate of capacity fade and impedance rise is about the same for particles 6 nm and larger, further showing that the size of the silicon nanoparticle is not predictive for calendar aging rates" (p. 303, Fig. 2d). Conclusions (p. 307): "There is no significant difference in calendar aging performance".

Mechanism: Thin-walled pressure-vessel argument (p. 304, eqn 1): "larger NPs, even if they are below the 150 nm limit for particle fracture, will still transmit more strain to the surrounding electrode matrix than smaller NPs". The ~150 nm fracture limit itself is cited from refs 9-11 (Introduction, p. 299).

Applicability: The paper studies 3-27 nm primary nanoparticles in ~89 wt% Si electrodes. Our measured particles are micron-scale (see paper_context.png) - roughly two orders of magnitude above the studied range, and also above the ~150 nm rupture threshold the paper cites. The paper's QUANTITATIVE results (e.g. 6-8 nm optimum) therefore do NOT apply to our particles. Only the qualitative trend is used here: within the silicon-bearing fraction, larger silicon units transmit more strain to the surrounding electrode and are expected to be worse for cycle life; the paper found no size effect on calendar life. Note also that our segmented objects may be agglomerates or embedded particles, not the primary crystallites the paper discusses, and our electrode is a different system (believed graphite bulk).

Risk ranking

No risk ranking is supported: the batches cannot be distinguished on silicon-particle size with these data. Even if they could, only the paper's qualitative cycle-life trend could be invoked, and the paper itself found no size effect on calendar life.

Limitations

- Material identity is assumed: bright BSE particles = silicon-based, grey bulk = graphite, from image appearance only (no EDS).
- A 2D polished section understates true 3D particle size (stereological bias); all D-values are section diameters.
- Pixel size (~25.0 nm/px) comes from TIFF resolution tags written by the acquisition software, not from a measured scale bar; a systematic calibration error would shift every size together.
- The image is the independent unit (n=7, 7, 17); Batch_3 has more images, so pairwise tests involving it have more power than the Batch_1-Batch_2 comparison.
- Two Batch_1 images (4ih2ggld, 5n1q8atc) returned roughly 2-4x the particle count of the rest at lower contrast; their thresholds dip into marginal bright material (flake interiors and rims), so their counts are inflated and sizes skewed small. Results are reported both with and without them.

- Segmented objects may be agglomerates or embedded composite particles, not primary crystallites - the feature the paper studies (3-27 nm primary particles) is far below our resolution and minimum object size.
- The paper's quantitative findings do not extend to micron-scale particles; only the qualitative larger-is-worse-for-cycling trend is invoked.